

The excess return for $t+1$ is the return of the fund at the end of the period, t to $t+1$, minus the benchmark portfolio formed using the weights at time t :

$$r_{t+1}^{ex} = r_{t+1} - \underbrace{[\beta_{SPY,t}SPY_{t+1} + \beta_{SPYV,t}SPYV_{t+1} + \beta_{SPYG,t}SPYG_{t+1}]}_{r_{t+1}^{bmk}}.$$

I graph the excess returns in Panel B of Figure 10.5. The cumulated excess returns are zero for LSV. Fidelity Magellan's returns trend downward (recall that Magellan significantly subtracts value in the full-sample regressions). Goldman's growth fund also has zero cumulative excess returns. The only fund with an upward trend is Berkshire Hathaway.

Style Analysis with Shorting

What if we allow shorting? In Figure 10.6, I allow the investor to take short positions in the ETFs. I use the following factor regression:

$$r_{t+1} - r_{f,t+1} = \alpha_{i,t} + \beta_{SPY,t}(SPY_{t+1} - r_{f,t+1}) + h_t(SPYV_{t+1} - SPYG_{t+1}) + \varepsilon_{t+1}. \quad (10.15)$$

This is the "ETF version" of the Fama–French (1993) regression that we estimated in equation (10.10), without the *SMB* factor, except that we allow the factor loadings to change over time. The *SPYV-SPYG* is an investment that goes long the value *SPYV* ETF and simultaneously shorts the growth *SPYG* ETF. Thus, it is analogous to the *HML* factor.

The factor loadings plotted in Panel A of Figure 10.6 show the strong value bias of LSV, with a positive h loading on the *SPYV-SPYG* factor. Magellan becomes more of a growth fund over time, with increasingly negative h loadings, as does Goldman's growth fund. Berkshire Hathaway's changing factor loadings from value to growth to value can be seen in its negative h loadings during 2008 and 2009.

Allowing shorting does not much change the cumulated excess returns in Panel B of Figure 10.6. But allowing shorting, not surprisingly, reduces the alphas. Magellan's trend line for cumulated excess returns becomes more negative when shorting is allowed. Although Buffett's excess returns are positive, they are shifted downward in Figure 10.6, Panel B, compared to the corresponding long-only picture in Figure 10.5, Panel B.

My final comment is that the problems of statistical inference with time-varying portfolio benchmarks are serious. It is hard enough to detect statistical significance with constant portfolio benchmarks, and estimated time-varying styles will have even larger standard errors.¹⁴

¹⁴ See comments by DiBartolomeo and Witkowski (1997).